

OAR FRAME ROWING SCHOOL

2-Day Student Lesson Plan

© Mac Brown Fly Fish · © Fly Fishing Guide School · Bryson City, NC | macbrownflyfish.com · flyfishingguideschool.com

14

SESSIONS

~9 hrs

ON-WATER TIME

5

CORE SKILL PILLARS

I-II

CLASS COVERAGE

Safety & PPE · Stroke Mechanics · Hydrology & River Reading · Navigation · Rescue

Resources: [Oar Frame Setup](#) · [Fly Casting Terms & Definitions](#) · [Fly Fishing Terms & Definitions](#)

QUICK REFERENCE GUIDE — INTERNATIONAL SCALE OF RIVER DIFFICULTY (ISRD)

CLASS	DESCRIPTION
Class I	Easy — fast water, small riffles, obvious channels. Self-rescue easy.
Class II	Novice — clear channels, some maneuvering, rare swimmer injury.
Class III	Intermediate — irregular waves, complex maneuvers, possible swamping. Reference only — not run in this program.
Class IV	Advanced — powerful, predictable water requiring precise boat control. Reference only — not run in this program.
Class V	Expert — violent, long, unavoidable drops. Rescue very difficult. Reference only — not run in this program.
Class VI	Extreme / unrunnable — risk of life even with full preparation. Reference only — not run in this program.

SIX CRITICAL RIVER HAZARDS

HAZARD	DESCRIPTION
Strainer	Debris or branches that water passes through but a body cannot.
Undercut	Rock eroded below water level — entrapment risk on impact.
Sieve	Constriction allowing water but not boats or bodies through.
Hydraulic / Hole	Water flows over obstacle, curls back upstream; can recirculate.
Foot Entrapment	Never stand in fast current — foot caught under rock.
Cold-Water Immersion	Raises effective difficulty of any class; impairs self-rescue.

RIVER FEATURES — READING WATER

FEATURE	DESCRIPTION
Downstream V / Tongue	Deepest, fastest channel into rapid — primary navigation target.
Upstream V	Surface indicator of a submerged obstacle — avoid.
Eddy	Calm reverse-flow pool behind obstacle — rest and staging zone.
Pillow / Cushion Wave	Foamy water on upstream face of rock — pressure wave.
Standing Wave	Stationary wave; surfable but can indicate deeper hazard.
Eddy Line	Boundary between main current and eddy — requires crossing angle.

W.O.R.M.S. SCOUTING FRAMEWORK

	DETAIL
W	Water
O	Obstacles

R	Route	Pick your line; mark entry, move, exit eddies
M	Management	Who is swimming? Who has throw bags? Safety positions?
S	Safety	Downstream hazards, bail-out eddies, rescue plan

Day 1 — Foundation

7:30 – 8:15 am Welcome, orientation & gear check

SAFETY

LEARNING OBJECTIVES

- Establish group norms and expectations
- Review participant skill levels
- Fit PFDs, helmets, and river footwear
- Introduce rescue equipment: throw bags, knives

KEY CONCEPTS

Type III / Type V PFD · Helmet fit · River knife placement · Whistle signals (1 = all stop, 3 = emergency)

ACTIVITIES

- Introductions and skill-level survey
- PFD fitting station — each student verified by instructor
- Safety briefing: signals, swim posture, group protocol

Checkpoint: Instructor sign-off: each student correctly fitted and able to demonstrate the 3-blast emergency whistle signal.

8:15 – 9:30 am Whitewater class system & hazard identification

THEORY

LEARNING OBJECTIVES

- Understand Class I–VI on the ISRD scale
- Recognize how flow, gradient, and obstacles combine
- Identify six critical river hazards
- Learn how water temperature elevates risk rating

KEY CONCEPTS

ISRD Class I–VI · Strainers · Undercuts · Sieves · Hydraulics · Foot entrapment · Cold-water immersion

ACTIVITIES

- Classroom: ISRD scale breakdown with photo examples of each class
- Hazard ID exercise: rank a photo set of river features by risk level
- Local river map study — identify today's run class and hazard zones

Checkpoint: Students correctly classify 8/10 rapid photos and identify 5 hazard types on the local map.

9:30 – 10:00 am · Break & gear to river

10:00 am – 12:00 pm Oar frame setup & stroke mechanics — flat water

STROKE SKILLS

LEARNING OBJECTIVES

- Set up and adjust an oar frame correctly
- Execute catch, drive, finish, and recovery with proper sequencing
- Understand the 1:2 drive-to-recovery ratio
- Develop independent oar control (pull vs. push)

KEY CONCEPTS

Oarlock over pivot point · Catch angle · Legs → back → arms sequence · Feathering on recovery · Forward vs. back stroke · One-oar rudder · Oar shipping safety

ACTIVITIES

- Oar frame rigging demo — students rig their own boats under supervision
- Flat-water drills: forward, back, and one-oar pivot on each side
- Paired feedback: one student rows, partner calls out sequencing errors
- Feathering drill — smooth oar rotation on each recovery pass

Checkpoint: Instructors observe drive sequence: legs → back → arms, with clean catches and no blade washing out.

12:00 – 12:45 pm · Lunch on the river

12:45 – 3:30 pm Reading water — hydrology & river features

RIVER READING

LEARNING OBJECTIVES

- Identify eddies, tongues, pillows, holes, and waves from the bank
- Understand laminar vs. turbulent flow and the formation of each feature
- Apply the W.O.R.M.S. scouting framework
- Pick a navigational line before entering the boat

KEY CONCEPTS

Downstream V / tongue · Eddy formation & eddy line · Pillow / cushion wave · Hydraulic recirculation · Standing waves · W.O.R.M.S. framework

ACTIVITIES

- Bankside feature walk: instructor points out each feature on live Class II water
- Students mark features on a river sketch before running
- Row Class II reach — enter and exit three separate eddies per student
- Debrief each rapid: was the read correct? What was unexpected?

Checkpoint: Each student successfully enters and exits three designated eddies using correct ferry angle, demonstrating tongue identification.

3:30 – 4:30 pm Day 1 debrief & self-assessment

REVIEW

LEARNING OBJECTIVES

- Consolidate the day's learning
- Identify personal skill gaps
- Preview Day 2 curriculum

ACTIVITIES

- Students complete written self-assessment on stroke mechanics and feature identification
- Group debrief: one thing learned, one thing harder than expected
- Instructor preview of Day 2 — navigation, ferries, rescue

Checkpoint: Completed self-assessment form collected; sets baseline for Day 2 instructor focus.

Day 2 — Application

8:00 – 8:45 am Morning review & swim safety / self-rescue

SAFETY

LEARNING OBJECTIVES

- Review defensive swim position
- Understand foot-entrapment hazard
- Know when to abandon the boat
- Distinguish active vs. passive self-rescue

KEY CONCEPTS

Defensive swim: feet up, toes up, downstream · Aggressive swim: angling toward shore · Foot entrapment — never stand in fast current · Hydraulic escape: dive deep, swim to side

ACTIVITIES

- Bankside demo of defensive swim — students practice on calm shallows
- Foot entrapment scenario: instructor demonstrates, students identify hazard
- Scenario Q&A: what do you do first after flipping in a hole on Class II water?

Checkpoint: Each student can verbally describe and physically demonstrate defensive swim posture and foot-entrapment awareness.

8:45 – 9:45 am Throw-bag rescue & shoreside rescue protocol

SAFETY

LEARNING OBJECTIVES

- Accurately deploy a throw bag to a swimmer
- Perform a belay hold without tying off
- Set up downstream safety positions
- Understand rescuer-first principle

KEY CONCEPTS

Throw bag — grab the rope, not the bag · Pendulum effect to shore · Belay position · Never tie rope to yourself · Two-rescuer assist (backup holder on PFD)

ACTIVITIES

- Throw-bag accuracy drill: hit a floating target from 30 ft, then 50 ft
- Live swimmer rescue pairs: one swims, one throws — rotate roles
- Set up a three-point downstream safety for a rapid — students assign positions

Checkpoint: Students land throw bag within arm's reach of swimmer in 2 of 3 attempts; demonstrate correct belay position.

9:45 – 10:00 am · Break & gear to river

10:00 am – 12:00 pm Ferry angles, pivots & eddy-hopping — Class I/II

NAVIGATION

LEARNING OBJECTIVES

- Execute upstream and downstream ferries with controlled angle
- Perform stern pivot to use current for turning
- Chain eddy entries and exits across a rapid
- Maintain situational awareness while maneuvering

KEY CONCEPTS

Upstream ferry: 45–90° angle, slower than current · Downstream ferry: pulling through laterals · Ferry angle vs. current speed tradeoff · Stern pivot — engage feature to spin bow · One-oar rudder for angle control

ACTIVITIES

- Ferry drill: cross river and land in target eddy — building toward confident Class II ferries
- Stern pivot practice: approach feature broadside, pull downstream oar to spin
- Eddy-hop run: students must touch 5 specified eddies on a Class I/II reach
- Instructor review of ferry angle between laps

Checkpoint: Students complete eddy-hop run with correct ferry technique in at least 4 of 5 target eddies.

12:00 – 12:45 pm · Lunch & scouting exercise at a Class II rapid

12:45 – 3:15 pm Navigating Class II — line selection & real-time adjustment

APPLIED NAV

LEARNING OBJECTIVES

- Apply all skills in a continuous Class II run
- Select and execute a planned line
- Make real-time corrections when off-line

- Develop calm decision-making under current pressure

KEY CONCEPTS

Pre-planned vs. reactive line selection · Reading moving water from the boat · Look-ahead: two moves ahead · Consequence evaluation: swim probability per line

ACTIVITIES

- Guided Class II run — instructor leads first lap, students follow planned lines
- Solo laps: each student runs 2 independent laps with safety in place
- Between laps: verbal debrief — where did you deviate and why?
- Optional: instructor photos key sections for stroke and line review

Checkpoint: Each student completes 2 solo laps with no unintended swims; instructor rates line selection quality on a 1–5 scale.

3:15 – 4:30 pm Final debrief, skills assessment & next steps

REVIEW

LEARNING OBJECTIVES

- Summarize two-day skill progression
- Issue individual written feedback
- Set a personal development plan

ACTIVITIES

- Written instructor assessment: stroke mechanics, river reading, safety, navigation
- Individual feedback session: strengths and one priority to develop next
- Program certificate and group photo at take-out

Checkpoint: Completed assessment form returned to each student; instructor retains copy for program records.